

Explaining the model

The 4 outer-fold models of repeat 1, reused for explainability and counterfactuals

Feature importance & SHAP

· the 4 saved fold models, refit

Refit with identical params, data and seed — reproduces the stored models and adds feature names

Feature Importance

get feature importance per
fold model → 4-model heatmap

SHAP

Explainer over each fold's own
test fold → mean |SHAP| heatmap

Counterfactuals (DiCE)

· the same 4 models, used as stored

Used exactly as stored — no refit. Each is wrapped with its own tuned threshold, so DiCE decides at that cut-off, not 0.5

Instances of interest — misclassified, or $|p - \text{threshold}| \leq 0.08$



DiCE genetic search — varies only the 6 actionable features



Outputs — local CFs + global CF importance